

Digispark Keyboard Scriptor Basic IDE

Input Prompting By: Mike Young – May 3, 2026

Using Gemini Pro 3.1

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YouTube: <https://youtu.be/hyzvSi9E6LM>

GitHub: <https://github.com/myoung8223/dks>

Introduction:

In this project we'll build a very basic, mostly portable (aside from a needed a driver installed), IDE (Integrated Development Environment) application for Windows that will allow the user to write new, open existing, save, and “transpile” keyboard scripts, similar to DuckyScript™, into Arduino ino files, compile, and upload those to a ATtiny85-based Digispark microcontroller, using arduino-cli. Such a device has usefulness in areas of automation and penetration testing.

Items needed:

- Windows PC
- [Tiny C Compiler](#)
- [Arduino CLI](#)
- [Digispark development board](#)

Initial Setup:

Start by creating a folder for this project. Call the folder “dks” for “Digispark Keyboard Scriptor”. Download the [Tiny C Compiler](#) and [Arduino CLI](#) zip files. Extract the Tiny C Compiler, “tcc”, into its own folder within the “dks” folder and move the extracted “arduino-cli.exe” file to the “dks” folder. Within the “tcc” folder you should see “tcc.exe”, the Tiny C Compiler main executable.

Create a text file in the “dks” folder called “arduino-cli.yaml”, with the following contents...

arduino-cli.yaml

```
directories:
  data: ./portable_data
  downloads: ./portable_data/staging
  user: ./portable_data/user
board_manager:
  additional_urls:
    - https://raw.githubusercontent.com/ArminJo/DigistumpArduino/master/package_digistump_index.json
```

That file basically tells Arduino CLI to save everything in “portable_data” in the “dks” folder, as opposed to saving needed files in the user profile. This helps make this solution semi-portable, aside from a needed driver to install for the Digispark so that Windows can interface with it.

Next, the toolchain needs to be initialized. Open a command prompt and change directory to the “dks” folder for this project, then run the following commands...

cmd commands

```
arduino-cli core update-index --config-file arduino-cli.yaml
arduino-cli core install digistump:avr --config-file arduino-cli.yaml
```

The first command “updates the index” and the second command installs the Digispark (Digistump) core. When installing the Digispark core you will likely encounter a notice that “Digistump LLC (usbser)” failed to install. That’s okay. Click on “Finish”.

The driver that fails to install is for Virtual COM ports (Serial). Because the Digispark doesn't have a dedicated hardware serial chip, it tries to “fake” one using software. Windows often blocks this specific old driver because it’s unsigned or conflicts with newer built-in USB-serial drivers. The libusb-win32 (Digispark/DigiUSB) drivers are what actually matter for our goal. Those allow the micronucleus uploader to talk to the chip and allow the PC to see it as a USB device.

Next, run this command to verify the installation...

cmd command

```
arduino-cli board listall digispark --config-file arduino-cli.yaml
```

You should see output like so...

cmd output

Board Name	FQBN
Digispark (16mhz - No USB)	digistump:avr:digispark-tiny16
Digispark (1mhz - No USB)	digistump:avr:digispark-tiny1
Digispark (8mhz - No USB)	digistump:avr:digispark-tiny8
Digispark (Default - 16.5mhz)	digistump:avr:digispark-tiny
Digispark Pro (16 Mhz) (32 byte buffer)	digistump:avr:digispark-pro32
Digispark Pro (16 Mhz) (64 byte buffer)	digistump:avr:digispark-pro64
Digispark Pro (Default 16 Mhz)	digistump:avr:digispark-pro

If you see that, the portable compiler is configured properly and ready for use.

Next, let’s create a “hello world” test to compile and upload to the Digispark. Create a folder in “dks” called “test_led”, and within that folder, create a text file called “test_led.ino”, containing the following text...

test_led.ino

```
void setup() { pinMode(1, OUTPUT); }
void loop() {
  digitalWrite(1, HIGH); delay(500);
  digitalWrite(1, LOW); delay(500);
}
```

Save that file and then run the following command...

cmd command

```
arduino-cli compile -b digistump:avr:digispark-tiny ./test_led --config-file arduino-cli.yaml
```

You should see output like so...

cmd output

```
Sketch uses 700 bytes (11%) of program storage space. Maximum is 6012 bytes.  
Global variables use 9 bytes of dynamic memory.
```

Now, let's upload the compiled "hello world" test program to the Digispark. Make sure your Digispark board is not plugged into your computer, then run this command...

cmd command

```
arduino-cli upload -b digistump:avr:digispark-tiny ./test_led --config-file arduino-cli.yaml
```

You may get a Windows Security notification asking if you wish to allow public and private networks to access the "mdns-discovery.exe" application. You can click either "Allow" or "Cancel" to that notice. That application is a utility the Arduino toolchain uses to find boards over a network (like an ESP32 or a Wi-Fi-enabled Arduino). Since the Digispark is strictly USB, that functionality isn't needed. Further below in this guide there is an option to completely suppress that issue from the start by replacing "mdns-discovery.exe" with a dummy program.

Watch the command prompt. It should state...

cmd output

```
Running Digispark Uploader...  
Plug in device now... (will timeout in 60 seconds)
```

Plug-in the Digispark board when you see that message. The program should then be uploaded and you should see output ending as follows...

cmd output

```
> Starting the user app ...  
running: 100% complete  
>> Micronucleus done. Thank you!  
New upload port: (default)
```

The board should then have a blinking red LED, in addition to its solid green power LED.

Windows GUI IDE Program

Next, we need to write the Windows GUI C program, to be compiled using tcc, that will handle allowing the user to write, open, save, save as, and build and upload the result to the Digispark.

Create a file called "dks.c", located in the "tcc" folder, with the following text content...

dks.c

```
// =====
// Digispark Keyboard Scripter IDE
// Copyright (c) 2026 Mike Young
// Licensed under the MIT License. See LICENSE file in the project root for full license information.
// =====

// =====
// Digispark Payload IDE & Transpiler
// =====

#include <windows.h>
#include <stdio.h>
#include <stdlib.h>
#include <ctype.h>
#include <string.h>

// =====
// --- MANUAL DEFINITIONS (Bypassing missing comdlg.h for TCC) ---
// =====
typedef struct {
    DWORD        lStructSize;
    HWND         hwndOwner;
    HINSTANCE    hInstance;
    LPCSTR       lpstrFilter;
    LPSTR        lpstrCustomFilter;
    DWORD        nMaxCustFilter;
    DWORD        nFilterIndex;
    LPSTR        lpstrFile;
    DWORD        nMaxFile;
    LPSTR        lpstrFileTitle;
    DWORD        nMaxFileTitle;
    LPCSTR       lpstrInitialDir;
    LPCSTR       lpstrTitle;
    DWORD        Flags;
    WORD         nFileOffset;
    WORD         nFileExtension;
    LPCSTR       lpstrDefExt;
    LPARAM       lCustData;
    void*        lpfnHook;
    LPCSTR       lpTemplateName;
} OPENFILENAME;

// Dialog Flags for File Open/Save behavior
#define OFN_HIDEREADONLY    0x00000004
#define OFN_OVERWRITEPROMPT 0x00000002
#define OFN_PATHMUSTEXIST   0x00000800
#define OFN_FILEMUSTEXIST   0x00001000
#define OFN_EXPLORER        0x00080000

// SS_NOPREFIX makes STATIC controls render '&' literally instead of
// treating it as a keyboard-mnemonic prefix. Guarded in case TCC omits it.
#ifndef SS_NOPREFIX
#define SS_NOPREFIX 0x00000080
#endif

// Tell the compiler these functions exist in the comdlg32 library
BOOL WINAPI GetOpenFileNameA(OPENFILENAME *lpofn);
BOOL WINAPI GetSaveFileNameA(OPENFILENAME *lpofn);
```

```

#define GetOpenFileName GetOpenFileNameA
#define GetSaveFileName GetSaveFileNameA

// ShellExecuteA lives in shell32; declare it manually in case the TCC winapi
// headers omit it (same approach used for the file-dialog functions above).
// NOTE: this requires linking shell32 -- add -lshell32 to the TCC command line.
HINSTANCE WINAPI ShellExecuteA(HWND, LPCSTR, LPCSTR, LPCSTR, LPCSTR, INT);
#ifdef ShellExecute
#define ShellExecute ShellExecuteA
#endif

// =====
// --- APPLICATION CONSTANTS & GLOBALS ---
// =====

// Menu Event IDs
#define IDM_NEW 1001
#define IDM_OPEN 1002
#define IDM_SAVE 1003
#define IDM_FLASH 1004
#define IDM_TOGGLE_DARK 1005
#define IDM_TOGGLE_WRAP 1006
#define IDM_SAVE_AS 1007
#define IDM_ABOUT 1008

// About-dialog control IDs
#define IDC_ABOUT_VISIT 2001
#define IDC_ABOUT_CLOSE 2002

// Application version string (shown in the About box; bump this on each release)
#define APP_VERSION "17"

// GitHub repository URL (shown in the About box and opened by the browser prompt)
#define GITHUB_URL "https://github.com/myoung8223/dks"

// USB HID Modifier Bitmasks (Used by DigiKeyboard)
#define MOD_CONTROL_LEFT (1 << 0)
#define MOD_SHIFT_LEFT (1 << 1)
#define MOD_ALT_LEFT (1 << 2)
#define MOD_GUI_LEFT (1 << 3)

// Global Handles
HWND hEdit; // The main text editor window
char currentFile[MAX_PATH] = ""; // Tracks the currently opened file path
char iniPath[MAX_PATH] = ""; // Tracks the path to the settings.ini file
HBRUSH hEditBkBrush = NULL; // Brush for Dark Mode background

int isDarkMode = 0; // Dark mode toggle state
int isWordWrap = 1; // Word wrap toggle state

// =====
// --- CONFIGURATION HELPERS ---
// =====

// Builds the absolute path to settings.ini in the same folder as the executable
void InitIniPath() {
    GetModuleFileName(NULL, iniPath, MAX_PATH);

```

```

    char *lastSlash = strrchr(iniPath, '\\');
    if (lastSlash) {
        strcpy(lastSlash + 1, "settings.ini");
    }
}

// Reads settings from the INI file
void LoadSettings() {
    isDarkMode = GetPrivateProfileInt("Settings", "DarkMode", 0, iniPath);
    isWordWrap = GetPrivateProfileInt("Settings", "WordWrap", 1, iniPath);
}

// Writes settings to the INI file
void SaveSettings() {
    char buffer[16];
    sprintf(buffer, "%d", isDarkMode);
    WritePrivateProfileString("Settings", "DarkMode", buffer, iniPath);
    sprintf(buffer, "%d", isWordWrap);
    WritePrivateProfileString("Settings", "WordWrap", buffer, iniPath);
}

// =====
// --- STRING UTILITIES ---
// =====

// Helper: Aggressive Trim
// Removes leading and trailing whitespace/newlines from a string.
char* clean_str(char *str) {
    char *end;
    // Trim leading space
    while(isspace((unsigned char)*str)) str++;
    if(*str == 0) return str; // All spaces?

    // Trim trailing space
    end = str + strlen(str) - 1;
    while(end > str && isspace((unsigned char)*end)) end--;
    end[1] = '\0';
    return str;
}

// Helper: Writes a C-string-escaped copy of src to the output file.
// Ensures backslashes, quotes, tabs, and newlines in a payload survive
// intact when embedded inside a "..." literal in the generated sketch.
void fputs_escaped(const char *src, FILE *out) {
    for (const char *p = src; *p != '\0'; p++) {
        switch (*p) {
            case '\\': fputs("\\\\", out); break;
            case '\"': fputs("\\\"", out); break;
            case '\n': fputs("\\n", out); break;
            case '\r': fputs("\\r", out); break;
            case '\t': fputs("\\t", out); break;
            default: fputc(*p, out); break;
        }
    }
}

// =====
// --- TRANSPILER ENGINE ---

```

```
// =====

// Converts the custom payload script into a valid Arduino C++ sketch.
// Returns 1 on success, 0 on syntax error or file error.
int transpile(const char* inputFilename) {
    int current_mod = 0;
    int default_delay = 0;

    FILE *in = fopen(inputFilename, "r");
    CreateDirectory("sketch", NULL);
    FILE *out = fopen("sketch\\sketch.ino", "w");

    if (!in) { if (out) fclose(out); return 0; }
    if (!out) { fclose(in); return 0; }

    // --- Write Arduino Sketch Header ---
    fprintf(out, "#include \"DigiKeyboard.h\"\n\n");

    // Inject USB HID key constants, but ONLY where the active core's library
    // doesn't already provide them. The original Digistump core omits these
    // (which is why they were injected); ArminJo's fork bundles a keylayouts.h
    // that defines them, so guarding with #ifndef defers to the library there
    // and avoids a wall of "redefined" warnings, while still supplying them on
    // cores that lack them.
    fprintf(out, "// --- USB HID Constants (defined only if the core omits them) ---\n");
    fprintf(out, "#ifndef KEY_ENTER\n#define KEY_ENTER 40\n#endif\n");
    fprintf(out, "#ifndef KEY_ESC\n#define KEY_ESC 41\n#endif\n");
    fprintf(out, "#ifndef KEY_ESCAPE\n#define KEY_ESCAPE 41\n#endif\n");
    fprintf(out, "#ifndef KEY_BACKSPACE\n#define KEY_BACKSPACE 42\n#endif\n");
    fprintf(out, "#ifndef KEY_TAB\n#define KEY_TAB 43\n#endif\n");
    fprintf(out, "#ifndef KEY_SPACE\n#define KEY_SPACE 44\n#endif\n");
    fprintf(out, "#ifndef KEY_MINUS\n#define KEY_MINUS 45\n#endif\n");
    fprintf(out, "#ifndef KEY_EQUAL\n#define KEY_EQUAL 46\n#endif\n");
    fprintf(out, "#ifndef KEY_DELETE\n#define KEY_DELETE 76\n#endif\n");
    fprintf(out, "#ifndef KEY_RIGHT\n#define KEY_RIGHT 79\n#endif\n");
    fprintf(out, "#ifndef KEY_LEFT\n#define KEY_LEFT 80\n#endif\n");
    fprintf(out, "#ifndef KEY_DOWN\n#define KEY_DOWN 81\n#endif\n");
    fprintf(out, "#ifndef KEY_UP\n#define KEY_UP 82\n#endif\n");

    // --- INJECT HELPER FUNCTIONS ---

    // Helper 1: Prevents dropped keystrokes by giving the OS time to process (PROGMEM optimized)
    fprintf(out, "// Helper: Rate-limits typing to prevent dropped keystrokes (PROGMEM optimized)\n");
    fprintf(out, "void safePrint(const __FlashStringHelper* text) {\n");
    fprintf(out, "    const char *p = (const char *)text;\n");
    fprintf(out, "    while (true) {\n");
    fprintf(out, "        unsigned char c = pgm_read_byte(p++);\n");
    fprintf(out, "        if (c == 0) break;\n");
    fprintf(out, "        DigiKeyboard.print((char)c);\n");
    fprintf(out, "        DigiKeyboard.delay(15);\n");
    fprintf(out, "    }\n");
    fprintf(out, "}\n");

    // Helper 2: Blinks N times, then pauses 2s, repeating until Pin 2 goes LOW
    fprintf(out, "// Helper: Blinks N times, then pauses 2s, repeating until Pin 2 goes LOW\n");
    fprintf(out, "void waitForButtonBlink(int count) {\n");
    fprintf(out, "    while(digitalRead(2) == HIGH) {\n");
    fprintf(out, "        for(int i=0; i < count; i++) {\n");
```

```

fprintf(out, "    digitalWrite(1, HIGH); DigiKeyboard.delay(200);\n");
fprintf(out, "    digitalWrite(1, LOW); DigiKeyboard.delay(200);\n");
fprintf(out, "    if(digitalRead(2) == LOW) return; // Exit if button pressed\n");
fprintf(out, "    }\n");
fprintf(out, "    // 2s pause, calling update() to keep USB alive\n");
fprintf(out, "    for(int j=0; j < 40; j++) {\n");
fprintf(out, "        DigiKeyboard.delay(50); DigiKeyboard.update();\n");
fprintf(out, "        if(digitalRead(2) == LOW) return;\n");
fprintf(out, "    }\n");
fprintf(out, "    }\n");
fprintf(out, "    DigiKeyboard.delay(200); // Debounce\n");
fprintf(out, "}\n\n");

// Start Arduino standard functions & initialize hardware pins
fprintf(out, "void setup() {\n");
fprintf(out, "    DigiKeyboard.sendKeyStroke(0); // Clear OS buffer\n");
fprintf(out, "    DigiKeyboard.delay(500); // Wait for OS to mount virtual USB\n");
fprintf(out, "    pinMode(1, OUTPUT); // Built-in LED\n");
fprintf(out, "    pinMode(2, INPUT_PULLUP); // Wait button (Pin 2 to GND)\n");
fprintf(out, "}\n\nvoid loop() {\n");
fprintf(out, "    DigiKeyboard.update();\n    DigiKeyboard.delay(3000);\n\n");

// --- Parse Input Script Line-by-Line ---
char line[256];
while (fgets(line, sizeof(line), in)) {
    char temp[256];
    strcpy(temp, line);

    // --- STRIP OUT COMMENTS (Quote-Aware) ---
    int in_quotes = 0;
    for (int i = 0; temp[i] != '\0'; i++) {
        if (temp[i] == '"') in_quotes = !in_quotes; // Toggle quote state
        if (temp[i] == '#' && !in_quotes) {
            temp[i] = '\0'; // Only terminate if NOT in a string
            break;
        }
    }
}

char *trimmed = clean_str(temp);
if (trimmed[0] == '\0') continue; // Skip empty lines

// --- CREATE AN ALL-CAPS VERSION FOR PARSING ---
char cmd[256];
strcpy(cmd, trimmed);
_strupr(cmd);

int is_keyboard_action = 0;

// Command: DEFAULTDELAY <ms>
if (strncmp(cmd, "DEFAULTDELAY", 12) == 0) {
    sscanf(cmd, "DEFAULTDELAY %d", &default_delay);
    continue;
}

// Command: string(...)
if (strncmp(cmd, "STRING(", 7) == 0) {
    // Search in 'trimmed' (the clean, comment-free version)
    char *start = strchr(trimmed, '"');

```



```

char *end = strrchr(trimmed, '\\');

if (start && end && start != end) {
    *end = '\\0';
    // RETROFIT: Use the new safePrint helper instead of DigiKeyboard.print.
    // Escape the payload so backslashes/quotes survive into the C++ literal.
    fputs("  safePrint(F(\"", out);
    fputs_escaped(start + 1, out);
    fputs("\\");\\n", out);
    is_keyboard_action = 1;
    *end = '\\'; // Restore it for consistency
} else {
    char errorMsg[512];
    sprintf(errorMsg, "Syntax error (missing quotes):\\n\\n%s", trimmed);
    MessageBox(NULL, errorMsg, "Transpiler Crash", MB_ICONERROR);
    fclose(in); fclose(out); return 0;
}
}

// Command: keydown(...)
else if (strncmp(cmd, "KEYDOWN(", 8) == 0) {
    char k1[32];
    if (sscanf(cmd, "KEYDOWN(%31[^)])", k1) >= 1) {
        char *ck1 = clean_str(k1);
        if (strcmp(ck1, "CONTROL") == 0)    current_mod |= MOD_CONTROL_LEFT;
        else if (strcmp(ck1, "ALT") == 0)   current_mod |= MOD_ALT_LEFT;
        else if (strcmp(ck1, "SHIFT") == 0) current_mod |= MOD_SHIFT_LEFT;
        else if (strcmp(ck1, "GUI") == 0)   current_mod |= MOD_GUI_LEFT;
        else {
            fprintf(out, "  DigiKeyboard.sendKeyPress(KEY_%s, %d);\\n", ck1, current_mod);
            is_keyboard_action = 0;
            goto trigger_delay;
        }
        fprintf(out, "  DigiKeyboard.sendKeyPress(0, %d);\\n", current_mod);
        is_keyboard_action = 0;
    }
}

// Command: keyup()
else if (strncmp(cmd, "KEYUP()", 7) == 0) {
    current_mod = 0;
    fprintf(out, "  DigiKeyboard.sendKeyPress(0, 0);\\n");
    is_keyboard_action = 1;
}

// Command: keys(...)
else if (strncmp(cmd, "KEYS(", 5) == 0) {
    char inner[128] = {0};
    if (sscanf(cmd, "KEYS(%127[^)])", inner) == 1) {
        char *t1 = strtok(inner, ",");
        char *t2 = strtok(NULL, ",");
        char *t3 = strtok(NULL, ",");

        if (t1) t1 = clean_str(t1);
        if (t2) t2 = clean_str(t2);
        if (t3) t3 = clean_str(t3);

        if (t1 && t2 && t3) {
            fprintf(out, "  DigiKeyboard.sendKeyPress(KEY_%s, MOD_%s_LEFT | MOD_%s_LEFT);\\n", t3,
t1, t2);

```

```

        fprintf(out, " DigiKeyboard.delay(100);\n");
        fprintf(out, " DigiKeyboard.sendKeyPress(0, 0);\n");
    } else if (t1 && t2) {
        fprintf(out, " DigiKeyboard.sendKeyStroke(KEY_%s, MOD_%s_LEFT);\n", t2, t1);
    } else if (t1) {
        fprintf(out, " DigiKeyboard.sendKeyStroke(KEY_%s);\n", t1);
    }
    is_keyboard_action = 1;
}
}

/*
// Command: key(...) [Legacy fallback]
else if (strncmp(cmd, "KEY(", 4) == 0) {
    char k1[32];
    if (sscanf(cmd, "KEY(%[^)])", k1) >= 1) {
        fprintf(out, " DigiKeyboard.sendKeyStroke(KEY_%s);\n", clean_str(k1));
        is_keyboard_action = 1;
    }
}

*/

// Command: key(...) [Legacy fallback]
else if (strncmp(cmd, "KEY(", 4) == 0) {
    char k1[32];
    if (sscanf(cmd, "KEY(%31[^)])", k1) >= 1) {
        if (current_mod > 0) {
            // Send the key WITH the held modifiers
            fprintf(out, " DigiKeyboard.sendKeyPress(KEY_%s, %d);\n", clean_str(k1), current_mod);
            fprintf(out, " DigiKeyboard.delay(50);\n"); // Brief tap
            // Release the key, but KEEP holding the modifiers
            fprintf(out, " DigiKeyboard.sendKeyPress(0, %d);\n", current_mod);
        } else {
            // Normal behavior if no modifiers are currently held
            fprintf(out, " DigiKeyboard.sendKeyStroke(KEY_%s);\n", clean_str(k1));
        }
        is_keyboard_action = 1;
    }
}

// Command: delay(<ms>)
else if (strncmp(cmd, "DELAY(", 6) == 0) {
    int ms;
    if (sscanf(cmd, "DELAY(%d)", &ms) >= 1) {
        fprintf(out, " DigiKeyboard.delay(%d);\n", ms);
    }
}

// Command: WAITBLINK(x)
else if (strncmp(cmd, "WAITBLINK(", 10) == 0) {
    int blinks = 0;
    if (sscanf(cmd, "WAITBLINK(%d)", &blinks) >= 1) {
        fprintf(out, " waitForButtonBlink(%d);\n", blinks);
    }
}

// Command: LED(ON)
else if (strcmp(cmd, "LED(ON)") == 0) {
    fprintf(out, " digitalWrite(1, HIGH);\n");
}

```

```

// Command: LED(OFF)
else if (strcmp(cmd, "LED(OFF)") == 0) {
    fprintf(out, "    digitalWrite(1, LOW);\n");
}
// Command: WAIT()
else if (strcmp(cmd, "WAIT()") == 0) {
    fprintf(out, "    while(digitalRead(2) == HIGH) { DigiKeyboard.update(); DigiKeyboard.delay(50);
}\n");
    fprintf(out, "    DigiKeyboard.delay(200);\n");
}
// CATCH-ALL: Syntax error / Unknown command!
else {
    char errorMsg[512];
    sprintf(errorMsg, "The transpiler encountered a syntax error and doesn't understand this
line:\n\n%s", trimmed);
    MessageBox(NULL, errorMsg, "Transpiler Error", MB_ICONERROR);

    fclose(in);
    fclose(out);
    return 0; // Abort build
}

// Apply DEFAULTDELAY if an actual keyboard action occurred
trigger_delay:
if (is_keyboard_action && default_delay > 0) {
    fprintf(out, "    DigiKeyboard.delay(%d); // Auto-delay\n", default_delay);
}
}

fprintf(out, "\n    for(;;){ DigiKeyboard.delay(1000); }\n\n");

fclose(in);
fclose(out);
return 1;
}

// =====
// --- FILE & PROCESS HELPERS ---
// =====

// Grabs all text from the Editor window and saves it to a file
void SaveFile(HWND hwnd, const char* filename) {
    int len = GetWindowTextLength(hEdit);
    if (len > 0) {
        char* buffer = (char*)malloc(len + 1);
        GetWindowText(hEdit, buffer, len + 1);
        FILE* fp = fopen(filename, "w");
        if (fp) {
            fputs(buffer, fp);
            fclose(fp);
        }
        free(buffer);
    } else {
        // Create empty file if no text
        FILE* fp = fopen(filename, "w");
        if (fp) fclose(fp);
    }
}

```

```

// Reads a file from disk and populates the Editor window
void LoadFile(HWND hwnd, const char* filename) {
    FILE* fp = fopen(filename, "rb");
    if (fp) {
        fseek(fp, 0, SEEK_END);
        long size = ftell(fp);
        rewind(fp);
        char* buffer = (char*)malloc(size + 1);
        fread(buffer, 1, size, fp);
        buffer[size] = '\0';
        fclose(fp);
        SetWindowText(hEdit, buffer);
        free(buffer);
    }
}

// Executes a CLI tool completely invisibly and waits for it to finish.
// Returns the exit code of the program (0 usually means success).
int RunCommandHidden(const char* cmd) {
    STARTUPINFO si;
    PROCESS_INFORMATION pi;

    ZeroMemory(&si, sizeof(si));
    si.cb = sizeof(si);
    si.dwFlags = STARTF_USESHOWWINDOW;
    si.wShowWindow = SW_HIDE; // Magic flag to hide the console window

    ZeroMemory(&pi, sizeof(pi));

    // CreateProcess requires a modifiable char buffer for the command
    char cmdBuffer[1024];
    strncpy(cmdBuffer, cmd, sizeof(cmdBuffer) - 1);
    cmdBuffer[sizeof(cmdBuffer) - 1] = '\0';

    // CREATE_NO_WINDOW prevents a console from even being allocated
    if (CreateProcess(NULL, cmdBuffer, NULL, NULL, FALSE, CREATE_NO_WINDOW, NULL, NULL, &si, &pi)) {
        WaitForSingleObject(pi.hProcess, INFINITE); // Block until complete

        DWORD exitCode;
        GetExitCodeProcess(pi.hProcess, &exitCode);

        CloseHandle(pi.hProcess);
        CloseHandle(pi.hThread);
        return (int)exitCode;
    }
    return -1; // Failed to launch process
}

// Copies the line starting at the first occurrence of 'needle' into 'dest',
// stopping at CR/LF. Returns 1 if found, 0 otherwise. Used to pull specific
// report lines (e.g. "Sketch uses ...") out of captured build output.
int extract_line(const char* src, const char* needle, char* dest, int destSize) {
    const char* p = strstr(src, needle);
    if (!p) { if (destSize > 0) dest[0] = '\0'; return 0; }
    int i = 0;
    while (p[i] != '\0' && p[i] != '\r' && p[i] != '\n' && i < destSize - 1) {
        dest[i] = p[i];
    }
}

```

```

        i++;
    }
    dest[i] = '\\0';
    return 1;
}

// Like RunCommandHidden, but also captures the child's stdout+stderr so the
// caller can read arduino-cli's messages (flash usage on success, linker
// errors on failure). Output is redirected to a temp file rather than an
// anonymous pipe to avoid the classic pipe-buffer deadlock on verbose builds.
// Returns the child's exit code, or -1 if the process could not be launched.
int RunCommandCapture(const char* cmd, char* outBuf, int outBufSize) {
    if (outBuf && outBufSize > 0) outBuf[0] = '\\0';

    // Build a unique temp file path for the child to write into
    char tmpDir[MAX_PATH], tmpFile[MAX_PATH];
    GetTempPath(MAX_PATH, tmpDir);
    if (!GetTempFileName(tmpDir, "dig", 0, tmpFile)) return -1;

    // The file handle must be inheritable so the child process can use it
    SECURITY_ATTRIBUTES sa;
    ZeroMemory(&sa, sizeof(sa));
    sa.nLength = sizeof(sa);
    sa.bInheritHandle = TRUE;

    HANDLE hFile = CreateFile(tmpFile, GENERIC_WRITE, FILE_SHARE_READ | FILE_SHARE_WRITE,
                             &sa, CREATE_ALWAYS, FILE_ATTRIBUTE_TEMPORARY, NULL);
    if (hFile == INVALID_HANDLE_VALUE) { DeleteFile(tmpFile); return -1; }

    STARTUPINFO si;
    PROCESS_INFORMATION pi;
    ZeroMemory(&si, sizeof(si));
    si.cb = sizeof(si);
    si.dwFlags = STARTF_USESHOWWINDOW | STARTF_USESTDHANDLES;
    si.wShowWindow = SW_HIDE;
    si.hStdInput = NULL; // arduino-cli compile/upload does not read stdin
    si.hStdOutput = hFile; // capture the flash-usage report
    si.hStdError = hFile; // capture linker/overflow errors too
    ZeroMemory(&pi, sizeof(pi));

    char cmdBuffer[1024];
    strncpy(cmdBuffer, cmd, sizeof(cmdBuffer) - 1);
    cmdBuffer[sizeof(cmdBuffer) - 1] = '\\0';

    int exitCode = -1;
    // NOTE: bInheritHandles MUST be TRUE here so the child inherits hFile
    if (CreateProcess(NULL, cmdBuffer, NULL, NULL, TRUE, CREATE_NO_WINDOW, NULL, NULL, &si, &pi)) {
        WaitForSingleObject(pi.hProcess, INFINITE);
        DWORD dwExit;
        GetExitCodeProcess(pi.hProcess, &dwExit);
        exitCode = (int)dwExit;
        CloseHandle(pi.hProcess);
        CloseHandle(pi.hThread);
    }

    CloseHandle(hFile); // release our write handle before reading the file back

    if (outBuf && outBufSize > 0) {

```

```

FILE* fp = fopen(tmpFile, "rb");
if (fp) {
    // The size report ("Sketch uses ...") is printed LAST, and a noisy
    // build can emit far more than outBufSize of warnings before it.
    // So keep the TAIL of the output: if the file is larger than the
    // buffer, seek to the final (outBufSize-1) bytes rather than the head.
    fseek(fp, 0, SEEK_END);
    long fsize = ftell(fp);
    long want = (long)outBufSize - 1;
    if (fsize > want) {
        fseek(fp, fsize - want, SEEK_SET);
    } else {
        rewind(fp);
    }
    size_t n = fread(outBuf, 1, (size_t)outBufSize - 1, fp);
    outBuf[n] = '\0';
    fclose(fp);
}

DeleteFile(tmpFile); // clean up the temp file
return exitCode;
}

// Recreates the Edit control to toggle styles like Word Wrap
void RecreateEditor(HWND hwndParent) {
    int len = GetWindowTextLength(hEdit);
    char* buffer = NULL;
    if (len > 0) {
        buffer = (char*)malloc(len + 1);
        GetWindowText(hEdit, buffer, len + 1);
    }

    HFONT hCurrentFont = (HFONT)SendMessage(hEdit, WM_GETFONT, 0, 0);

    DWORD editStyle = WS_CHILD | WS_VISIBLE | WS_VSCROLL | ES_MULTILINE | ES_AUTOVSCROLL | ES_WANTRETURN;
    if (!isWordWrap) editStyle |= ES_AUTOHSCROLL | WS_HSCROLL;

    HWND hOld = hEdit;
    hEdit = CreateWindowEx(WS_EX_CLIENTEDGE, "EDIT", buffer ? buffer : "", editStyle,
        0, 0, 0, 0, hwndParent, NULL, GetModuleHandle(NULL), NULL);

    SendMessage(hEdit, WM_SETFONT, (WPARAM)hCurrentFont, TRUE);
    DestroyWindow(hOld);

    RECT rc; GetClientRect(hwndParent, &rc);
    MoveWindow(hEdit, 0, 0, rc.right, rc.bottom, TRUE);
    if (buffer) free(buffer);
}

// =====
// --- ABOUT DIALOG (custom window, so we control button labels & the chime) ---
// =====

// A standard MessageBox can't relabel its buttons, and its MB_ICON* flags play
// a system sound on open. So the About box is a small custom window instead:
// arbitrary button text ("Visit GitHub Link" / "Close") and no sound.
LRESULT CALLBACK AboutWndProc(HWND hDlg, UINT msg, WPARAM wParam, LPARAM lParam) {

```

```

switch (msg) {
    case WM_CREATE: {
        // Fonts are created once and reused for the life of the process.
        // Segoe UI gives a modern look versus the dated default GUI font.
        static HFONT hTitleFont = NULL, hBodyFont = NULL;
        if (!hBodyFont)
            hBodyFont = CreateFont(-13, 0, 0, 0, FW_NORMAL, FALSE, FALSE, FALSE,
                                   ANSI_CHARSET, OUT_DEFAULT_PRECIS, CLIP_DEFAULT_PRECIS,
                                   DEFAULT_QUALITY, DEFAULT_PITCH | FF_SWISS, "Segoe UI");
        if (!hTitleFont)
            hTitleFont = CreateFont(-19, 0, 0, 0, FW_BOLD, FALSE, FALSE, FALSE,
                                   ANSI_CHARSET, OUT_DEFAULT_PRECIS, CLIP_DEFAULT_PRECIS,
                                   DEFAULT_QUALITY, DEFAULT_PITCH | FF_SWISS, "Segoe UI");

        // Bold title on its own line
        HWND hTitle = CreateWindow("STATIC",
                                   "Digispark Keyboard Scripter IDE",
                                   WS_CHILD | WS_VISIBLE | SS_LEFT | SS_NOPREFIX,
                                   22, 18, 406, 26, hDlg, NULL, GetModuleHandle(NULL), NULL);
        SendMessage(hTitle, WM_SETFONT, (WPARAM)hTitleFont, TRUE);

        // Body text. SS_NOPREFIX makes '&' render literally (no mnemonic underline).
        HWND hText = CreateWindow("STATIC",
                                   "Version " APP_VERSION "\n\n"
                                   "Creator & Project Lead:\n"
                                   "    Mike Young\n\n"
                                   "Software Engineering:\n"
                                   "    Google Gemini and Anthropic Claude\n\n"
                                   "GitHub repository:\n"
                                   "    " GITHUB_URL,
                                   WS_CHILD | WS_VISIBLE | SS_LEFT | SS_NOPREFIX,
                                   22, 50, 406, 188, hDlg, NULL, GetModuleHandle(NULL), NULL);
        SendMessage(hText, WM_SETFONT, (WPARAM)hBodyFont, TRUE);

        HWND hVisit = CreateWindow("BUTTON", "Visit GitHub Link",
                                   WS_CHILD | WS_VISIBLE | BS_PUSHBUTTON,
                                   22, 255, 150, 30, hDlg, (HMENU)(UINT_PTR)IDC_ABOUT_VISIT,
                                   GetModuleHandle(NULL), NULL);
        SendMessage(hVisit, WM_SETFONT, (WPARAM)hBodyFont, TRUE);

        HWND hClose = CreateWindow("BUTTON", "Close",
                                   WS_CHILD | WS_VISIBLE | BS_PUSHBUTTON,
                                   333, 255, 95, 30, hDlg, (HMENU)(UINT_PTR)IDC_ABOUT_CLOSE,
                                   GetModuleHandle(NULL), NULL);
        SendMessage(hClose, WM_SETFONT, (WPARAM)hBodyFont, TRUE);
        break;
    }

    case WM_COMMAND:
        switch (LOWORD(wParam)) {
            case IDC_ABOUT_VISIT:
                ShellExecute(hDlg, "open", GITHUB_URL, NULL, NULL, SW_SHOWNORMAL);
                break;
            case IDC_ABOUT_CLOSE:
                DestroyWindow(hDlg);
                break;
        }
        break;
}

```

```

        case WM_CLOSE:
            DestroyWindow(hDlg); // Close button / title-bar X -- must NOT quit the app
            break;

        default:
            return DefWindowProc(hDlg, msg, wParam, lParam);
    }
    return 0;
}

// Creates and shows the About dialog, centered over the parent window.
void ShowAboutDialog(HWND hParent) {
    int w = 450, h = 340;
    RECT rp;
    GetWindowRect(hParent, &rp);
    int x = rp.left + ((rp.right - rp.left) - w) / 2;
    int y = rp.top + ((rp.bottom - rp.top) - h) / 2;
    if (x < 0) x = 0;
    if (y < 0) y = 0;

    CreateWindowEx(WS_EX_DLGMODALFRAME, "AboutDlgClass",
        "About Digispark Keyboard Scripter IDE",
        WS_POPUP | WS_CAPTION | WS_SYSMENU | WS_VISIBLE,
        x, y, w, h, hParent, NULL, GetModuleHandle(NULL), NULL);
}

// =====
// --- WINDOWS GUI EVENT LOOP ---
// =====

LRESULT CALLBACK WndProc(HWND hwnd, UINT msg, WPARAM wParam, LPARAM lParam) {
    switch (msg) {
        case WM_CREATE: {
            // 1. Build the Application Menus
            HMENU hMenu = CreateMenu();
            HMENU hFileMenu = CreatePopupMenu();
            HMENU hBuildMenu = CreatePopupMenu();
            HMENU hOptionsMenu = CreatePopupMenu();

            AppendMenu(hFileMenu, MF_STRING, IDM_NEW, "New");
            AppendMenu(hFileMenu, MF_STRING, IDM_OPEN, "Open...");
            AppendMenu(hFileMenu, MF_STRING, IDM_SAVE, "Save");
            AppendMenu(hFileMenu, MF_STRING, IDM_SAVE_AS, "Save As...");
            AppendMenu(hMenu, MF_POPUP, (UINT_PTR)hFileMenu, "File");

            AppendMenu(hBuildMenu, MF_STRING, IDM_FLASH, "Compile && Flash Script");
            AppendMenu(hMenu, MF_POPUP, (UINT_PTR)hBuildMenu, "Build");

            AppendMenu(hOptionsMenu, MF_STRING | (isDarkMode ? MF_CHECKED : 0), IDM_TOGGLE_DARK, "Dark
Mode");
            AppendMenu(hOptionsMenu, MF_STRING | (isWordWrap ? MF_CHECKED : 0), IDM_TOGGLE_WRAP, "Word
Wrap");

            AppendMenu(hOptionsMenu, MF_SEPARATOR, 0, NULL);
            AppendMenu(hOptionsMenu, MF_STRING, IDM_ABOUT, "About...");
            AppendMenu(hMenu, MF_POPUP, (UINT_PTR)hOptionsMenu, "Options");

            SetMenu(hwnd, hMenu);
        }
    }
}

```



```

// 2. Create the Text Editor Control
DWORD editStyle = WS_CHILD | WS_VISIBLE | WS_VSCROLL | ES_MULTILINE | ES_AUTOVSCROLL |
ES_WANTRETURN;
if (!isWordWrap) editStyle |= ES_AUTOHSCROLL | WS_HSCROLL;

hEdit = CreateWindowEx(WS_EX_CLIENTEDGE, "EDIT", "",
    editStyle,
    0, 0, 0, 0, hwnd, NULL, GetModuleHandle(NULL), NULL);

// Brush for Dark Mode
hEditBkBrush = CreateSolidBrush(RGB(30, 30, 30));

// 3. Set a monospaced font for coding
HFONT hFont = CreateFont(18, 0, 0, 0, FW_NORMAL, FALSE, FALSE, FALSE, ANSI_CHARSET,
    OUT_DEFAULT_PRECIS, CLIP_DEFAULT_PRECIS, DEFAULT_QUALITY,
    DEFAULT_PITCH | FF_SWISS, "Consolas");
SendMessage(hEdit, WM_SETFONT, (WPARAM)hFont, TRUE);
break;
}

case WM_CTLCOLOREDIT: {
    HDC hdc = (HDC)wParam;
    if (isDarkMode) {
        SetTextColor(hdc, RGB(220, 220, 220)); // Off-white text
        SetBkColor(hdc, RGB(30, 30, 30));      // Dark gray background
        return (INT_PTR)hEditBkBrush;
    } else {
        SetTextColor(hdc, RGB(0, 0, 0));        // Black text
        SetBkColor(hdc, RGB(255, 255, 255));    // White background
        return (INT_PTR)GetStockObject(WHITE_BRUSH);
    }
}

case WM_SIZE: {
    // Automatically resize the text editor to fill the parent window
    MoveWindow(hEdit, 0, 0, LOWORD(lParam), HIWORD(lParam), TRUE);
    break;
}

case WM_COMMAND: {
    // Handle Menu Clicks
    switch (LOWORD(wParam)) {

        case IDM_NEW:
            SetWindowText(hEdit, "");
            strcpy(currentFile, "");
            SetWindowText(hwnd, "Digispark Keyboard Scripter IDE - Untitled");
            break;

        case IDM_OPEN: {
            OPENFILENAME ofn;
            ZeroMemory(&ofn, sizeof(ofn));
            ofn.lStructSize = sizeof(ofn);
            ofn.hwndOwner = hwnd;
            ofn.lpstrFilter = "Text Files (*.txt)\\0*.txt\\0All Files (*.*)\\0*.*\\0";
            ofn.lpstrFile = currentFile;
            ofn.nMaxFile = MAX_PATH;

```

```

        ofn.Flags = OFN_EXPLORER | OFN_FILEMUSTEXIST;

        if (GetOpenFileName(&ofn)) {
            LoadFile(hwnd, currentFile);
            char title[MAX_PATH + 64];
            sprintf(title, sizeof(title), "Digispark Keyboard Scriptor IDE - %s",
currentFile);
            SetWindowText(hwnd, title);
        }
        break;
    }

    case IDM_SAVE: {
        if (strlen(currentFile) > 0) {
            SaveFile(hwnd, currentFile);
        } else {
            // If there is no file opened/saved yet, trigger Save As logic
            SendMessage(hwnd, WM_COMMAND, IDM_SAVE_AS, 0);
        }
        break;
    }

    case IDM_SAVE_AS: {
        OPENFILENAME ofn;
        ZeroMemory(&ofn, sizeof(ofn));
        ofn.lStructSize = sizeof(ofn);
        ofn.hwndOwner = hwnd;
        ofn.lpstrFilter = "Text Files (*.txt)\0*.txt\0All Files (*.*)\0*.*\0";
        ofn.lpstrFile = currentFile;
        ofn.nMaxFile = MAX_PATH;
        ofn.Flags = OFN_EXPLORER | OFN_PATHMUSTEXIST | OFN_HIDEREADONLY | OFN_OVERWRITEPROMPT;
        ofn.lpstrDefExt = "txt";

        if (GetSaveFileName(&ofn)) {
            SaveFile(hwnd, currentFile);
            char title[MAX_PATH + 64];
            sprintf(title, sizeof(title), "Digispark Keyboard Scriptor IDE - %s",
currentFile);
            SetWindowText(hwnd, title);
        }
        break;
    }

    case IDM_TOGGLE_DARK:
        isDarkMode = !isDarkMode;
        CheckMenuItem(GetMenu(hwnd), IDM_TOGGLE_DARK, isDarkMode ? MF_CHECKED : MF_UNCHECKED);
        InvalidateRect(hEdit, NULL, TRUE);
        break;

    case IDM_TOGGLE_WRAP:
        isWordWrap = !isWordWrap;
        CheckMenuItem(GetMenu(hwnd), IDM_TOGGLE_WRAP, isWordWrap ? MF_CHECKED : MF_UNCHECKED);
        RecreateEditor(hwnd);
        break;

    case IDM_ABOUT:
        ShowAboutDialog(hwnd);
        break;

```

```

case IDM_FLASH: {
    // Step A: Save current editor text to a temp file
    SaveFile(hwnd, "temp_payload.txt");

    // Step B: Transpile script to Arduino sketch
    if (transpile("temp_payload.txt")) {

        // Step C: Compile sketch via arduino-cli, capturing its output
        char buildOut[8192];
        int compileStatus = RunCommandCapture(
            "arduino-cli compile --config-file arduino-cli.yaml --fqbn
digistump:avr:digispark-tiny sketch",
            buildOut, sizeof(buildOut));

        if (compileStatus != 0) {
            // Distinguish "too big for flash" from an ordinary syntax error.
            // Linker wording varies by binutils version, so match a few phrases.
            int tooBig = (strstr(buildOut, "overflowed") != NULL)
                || (strstr(buildOut, "will not fit in region") != NULL)
                || (strstr(buildOut, "not within region") != NULL);

            char errMsg[8192 + 512];
            if (tooBig) {
                snprintf(errMsg, sizeof(errMsg),
                    "Compilation failed: the sketch is TOO LARGE for the Digispark's
flash.\n\n"

                    "Shorten the script, or move long payloads to a downloaded stage.\n\n"
                    "--- Compiler output ---\n%s", buildOut);
            } else {
                snprintf(errMsg, sizeof(errMsg),
                    "Compilation failed!\n\nThere is likely a syntax error in your
script.\n\n"

                    "--- Compiler output ---\n%s", buildOut);
            }
            MessageBox(hwnd, errMsg, "Build Error", MB_ICONERROR);
            break;
        }

        // Step D: Report flash/RAM usage, then inform user to plug in hardware
        char sketchLine[256] = "", ramLine[256] = "";
        extract_line(buildOut, "Sketch uses", sketchLine, sizeof(sketchLine));
        extract_line(buildOut, "Global variables", ramLine, sizeof(ramLine));

        char okMsg[1024];
        snprintf(okMsg, sizeof(okMsg),
            "Compilation successful!\n\n%s\n%s\n\n"
            "Unplug your Digispark, click OK, and then plug it back in within 60 seconds to
start flashing.",
            sketchLine[0] ? sketchLine : "(flash usage unavailable)",
            ramLine[0] ? ramLine : "");
        MessageBox(hwnd, okMsg, "Ready to Flash", MB_OK | MB_ICONINFORMATION);

        // Step E: Upload firmware via arduino-cli (Micronucleus)
        int uploadStatus = RunCommandHidden("arduino-cli upload --config-file
arduino-cli.yaml --fqbn digistump:avr:digispark-tiny sketch");

        if (uploadStatus != 0) {

```

```

        MessageBox(hwnd, "Flashing failed! Did the Digispark time out?", "Flashing
Failed", MB_ICONERROR);
    } else {
        MessageBox(hwnd, "Script flashed successfully!", "Flashing Succeeded", MB_OK);
    }

    } else {
        MessageBox(hwnd, "Transpilation failed. Check your script for invalid commands.",
"Transpilation Failed", MB_ICONERROR);
    }
    break;
}
}
break;
}

case WM_DESTROY:
    SaveSettings(); // Save user preferences before closing
    if (hEditBkBrush) DeleteObject(hEditBkBrush); // Clean up GDI object
    PostQuitMessage(0); // Exit application safely
    break;

default:
    return DefWindowProc(hwnd, msg, wParam, lParam);
}
return 0;
}

// =====
// --- APPLICATION ENTRY POINT ---
// =====
int WINAPI WinMain(HINSTANCE hInstance, HINSTANCE hPrevInstance, LPSTR lpCmdLine, int nCmdShow) {
    // Initialize file path and load preferences before building the UI
    InitIniPath();
    LoadSettings();

    WNDCLASS wc = {0};
    wc.lpfnWndProc = WndProc;
    wc.hInstance = hInstance;
    wc.lpszClassName = "DigiIDEClass";
    wc.hCursor = LoadCursor(NULL, IDC_ARROW);
    wc.hbrBackground = (HBRUSH)(COLOR_WINDOW + 1);

    RegisterClass(&wc);

    // Register the About dialog's window class
    WNDCLASS wcAbout = {0};
    wcAbout.lpfnWndProc = AboutWndProc;
    wcAbout.hInstance = hInstance;
    wcAbout.lpszClassName = "AboutDlgClass";
    wcAbout.hCursor = LoadCursor(NULL, IDC_ARROW);
    wcAbout.hbrBackground = (HBRUSH)(COLOR_BTNFACE + 1);
    RegisterClass(&wcAbout);

    // Create the Main Window
    HWND hwnd = CreateWindowEx(0, "DigiIDEClass", "Digispark Keyboard Scriptor IDE - Untitled",
        WS_OVERLAPPEDWINDOW, CW_USEDEFAULT, CW_USEDEFAULT, 800, 600,
        NULL, NULL, hInstance, NULL);

```

```

ShowWindow(hwnd, nCmdShow);

// Main Message Pump
MSG msg;
while (GetMessage(&msg, NULL, 0, 0)) {
    TranslateMessage(&msg);
    DispatchMessage(&msg);
}
return msg.wParam;
}

```

Save that file. Note that this is the LICENSE file...

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Go to the command prompt, make sure you are in the "tcc" folder, then run this command to compile that C source code file into a Windows GUI application...

cmd command

```
tcc dks.c -o dks.exe -luser32 -lgdi32 -lcomdlg32 -lshell32 -Wl,-subsystem=windows
```

That command should succeed and generate the file "dks.exe" in the "tcc" folder. Run this command to move the "dks.exe" file to the parent folder, so it can run alongside "arduino-cli.exe"...

cmd command

```
move dks.exe ..
```

Next, launch “dks.exe” from the “dks” folder, then copy and paste the following sample script into the editor and save it as “payload.txt”...

payload.txt

```
defaultdelay 300    # delay 300ms between commands

waitblink(1)    # blink the red LED once every 2s and wait for button press
keys(gui, r)      # press [Windows] + [R]
string("notepad") # type "notepad"
key(enter)        # press [Enter]
delay(1000)       # wait 1 second for Notepad to appear
string("Hello, world!") # type "Hello, world!"
led(on)          # turn on red LED to indicate script is done running
```

Click on **Build** → **Compile & Flash Script**. The script will be transpiled into an ino file. You should get a notification dialog window stating **Compilation successful!** Unplug the Digispark, click OK to the notification, then re-insert the Digispark. The system should proceed with flashing the compiled ino to the Digispark. When it finishes you should see another notification dialog window appear stating **Script flashed successfully!** Click **OK**. The payload should then run.

Note that error checking of the script is very basic. A simple syntax error in your keyboard script might still compile and result in unpredictable or omitted output. In some cases, severe syntax errors may cause the generated ino file to not compile, resulting in an error notification. Note too that you can debug your script by opening and analyzing the generated ino file “sketch.ino”, located in the “sketch” folder.

Semi-Portable Use

The generated application isn’t completely portable, as the Digistump drivers need to be installed on any Windows computer that will be used for uploading compiled scripts. Additionally, regarding the “mdns-discovery.exe” issue, while that’s just a one-time issue, as firewall rules get saved when you make a choice when prompted about that executable, that problem can be suppressed from the beginning by replacing “mdns-discovery.exe” with a dummy version of the program that does nothing.

Start by creating a dummy program called “dummy.c” with the following content, saved to the tcc directory...

dummy.c

```
#include <windows.h>

int WINAPI WinMain(HINSTANCE hInstance, HINSTANCE hPrevInstance, LPSTR lpCmdLine, int nCmdShow) {
    return 0;    // do absolutely nothing and exit successfully
}
```

In the command prompt, change directory to the tcc directory and compile that program with this command...

cmd command

```
tcc dummy.c -o mdns-discovery.exe
```

Move the generated dummy “mdns-discovery.exe” file to...

```
...\dks\portable_data\packages\builtin\tools\mdns-discovery\1.0.12
```

When prompted to replace it, proceed with doing so.

Create a batch file named “Install Digistump Drivers.cmd” that resides in the “dks” folder, with the following content...

```
"Install Digistump Drivers.cmd"
```

```
@echo off
"%~dp0portable_data\packages\digistump\tools\micronucleus\2.0a4\install.exe"
```

The contents of the “dks” folder can now be compressed into a zip file, copied to another system, extracted, and all that needs to be performed is to install the Digistump drivers. Installing the Digistump drivers can be performed by running the installer, which is located here...

```
...\dks\portable_data\packages\digistump\tools\micronucleus\2.0a4\install.exe
```

Upon doing that, you should be able to run “dks.exe” and proceed with writing, compiling, and uploading scripts to the Digispark. Though instead of navigating to that folder, just run the “Install Digistump Drivers.cmd” batch file we just created, from the “dks” folder.

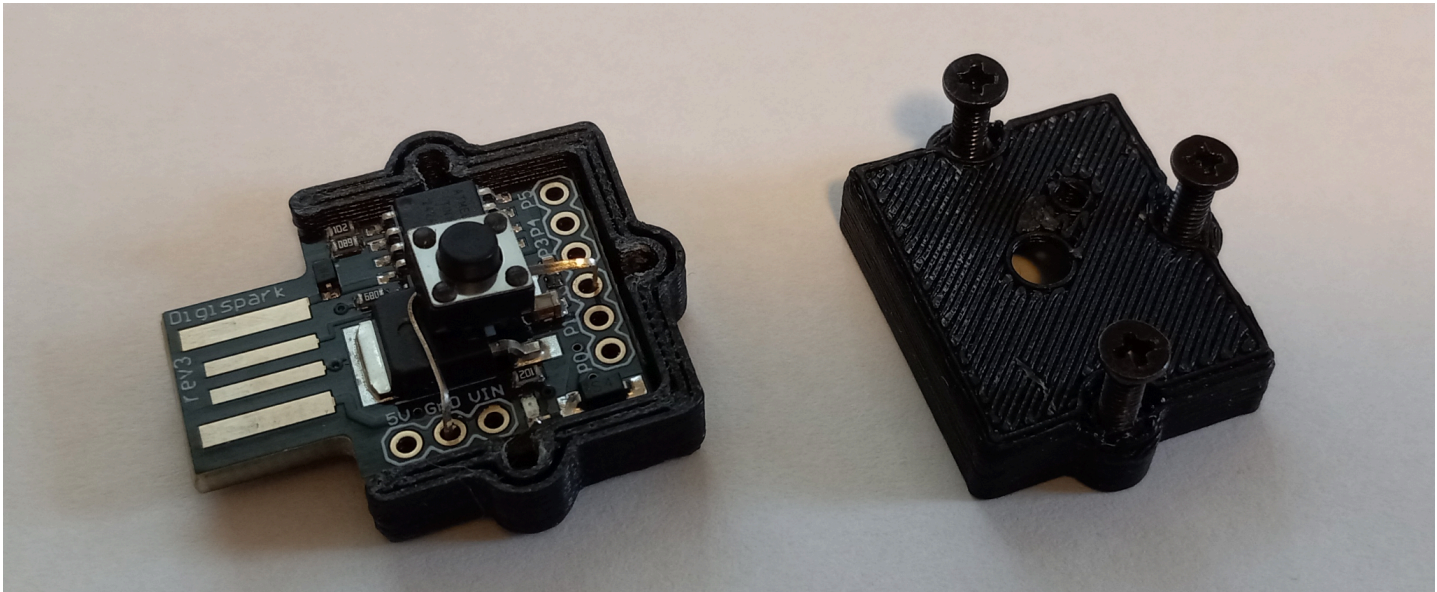
Note that the command to forget about Windows Firewall settings for “mdns-discovery.exe” is (replace **PATH** according to your specific case)...

```
cmd command
```

```
netsh advfirewall firewall delete rule name=all
program="PATH\dks\portable_data\packages\builtin\tools\mdns-discovery\1.0.12\mdns-discovery.exe"
```

Additional Hardware Setup

Print the case pieces. Connect a 6x6x4mm tactile pushbutton switch (through-hole, SPST-NO) with long legs to GND and P2 on the Digispark. In my case, I just bend the pins into the GPIO holes and that seems to work reliably due to the friction fit and compression of the case, but applying some solder may be advisable. Position the switch center of the board so that it will fit into the cavity of the top portion of the case. Fit the case pieces together, compress them with your fingers, and screw-in the three M2.5x8mm case screws.



Specifications for Keyboard Scripting

This language automates keystrokes and hardware states on a Digispark ATtiny85. It is line-based, **case-insensitive** for command names and key identifiers, supports inline comments, and allows up to three simultaneous key presses.

1. Core Commands

Command	Description	Example
<code>string("text")</code>	Types the literal string provided inside the quotes. (Case-sensitive inside quotes)	<code>string("notepad.exe")</code>
<code>keys(args...)</code>	The universal key command. Supports 1, 2, or 3 arguments (modifiers first, key last).	<code>keys(control, alt, delete)</code>
<code>key(key)</code>	An alias for <code>keys(key)</code> . Presses and releases a single key.	<code>key(enter)</code>
<code>keydown(key/mod)</code>	Presses and holds a key or modifier. Modifiers "stack" instantly.	<code>keydown(control)</code>
<code>keyup()</code>	Releases all currently held keys and modifiers instantly.	<code>keyup()</code>
<code>delay(ms)</code>	Pauses execution for the specified number of milliseconds.	<code>delay(1000)</code>
<code>defaultdelay ms</code>	Sets a global delay (in ms) added after every keyboard action. Increase this value slightly if keypresses are being missed or mistyped.	<code>defaultdelay 200</code>
<code>led(on)</code>	Turns on the Digispark's built-in status LED (Pin 1).	<code>led(on)</code>

<code>led(off)</code>	Turns off the Digispark's built-in status LED.	<code>led(off)</code>
<code>wait()</code>	Pauses the script infinitely until the physical button (wired to Pin 2 and GND) is pressed.	<code>wait()</code>
<code>waitblink(x)</code>	Pauses the script infinitely until the physical button (wired to Pin 2 and GND) is pressed, while blinking x times, pausing 2 seconds, and repeating.	<code>waitblink(3)</code>

2. Modifier Keys (MOD)

These are used within `keys()` and `keydown()` functions. They must be placed before the primary key in a multi-key command.

- **GUI** (Windows / Command Key)
- **CONTROL**
- **SHIFT**
- **ALT**

3. Key Definitions (KEY_NAME)

Common identifiers include:

- **Alpha-numeric:** A through Z, 0 through 9
- **Navigation:** UP, DOWN, LEFT, RIGHT, PAGE_UP, PAGE_DOWN, HOME, END
- **Editing:** ENTER, BACKSPACE, DELETE, TAB, SPACE, ESC (or ESCAPE)
- **Function Keys:** F1 through F12
- **System:** PRINTSCREEN, SCROLL_LOCK, PAUSE

4. Scripting Logic & Rules

- **Case Insensitivity:** Commands and key identifiers can be written in uppercase, lowercase, or mixed case (e.g., `DELAY(100)`, `delay(100)`, and `Delay(100)` are identical). **Exception:** Text written inside the quotes of a `string("...")` command will be typed exactly as formatted.
 - **Comments:** Use the `#` symbol to add comments. The transpiler will ignore anything on a line after the `#` symbol.
 - **Argument Order:** When using multiple keys in `keys()`, the modifiers must come first and the primary key must be last.
 -  **Valid:** `keys(CONTROL, ALT, DELETE)`
 -  **Invalid:** `keys(DELETE, CONTROL, ALT)`
 - **Initialization:** Every script starts with an automatic 3-second delay to allow the OS to mount the USB drivers.
 - **Stateful Modifiers:** If you use `keydown(SHIFT)`, the Shift key remains held. All subsequent `string()` or `keys()` commands will be typed as if Shift is held until you call `keyup()`.
 - **Hardware Hold:** The 3-key version of `keys()` uses a 100ms hardware-level hold to ensure the command is registered by system-level security screens (like the Windows Lock screen).
-

5. Example Payloads

Example 1: Open Notepad and Type

```
defaultdelay 250

keys(gui, r)      # open Run dialog
string("notepad")
key(enter)
delay(1000)      # wait for Notepad to open
string("This payload was deployed via Digispark!")
key(enter)
```

Example 2: Selection & Formatting (Stateful Keys) This script types text, then uses `keydown` to hold modifiers while navigating, mimicking a user highlighting and copying text.

```
defaultdelay 150

string("Automating with Digispark")
key(enter)

# highlight the line above
keydown(shift)
keys(up)
keyup()

# copy and paste
keys(control, C)
keys(right)
keys(enter)
keys(control, v)

# system command
keys(control, alt, delete)
```

Example 3: Hardware Interaction (Wait for User) This script waits for the user to push a physical button before executing the payload, using the LED to indicate when it is ready.

```
led(on)    # turn on the LED to signal the USB is mounted and ready
wait()     # pause script execution until the pin 2 button is pressed
led(off)   # turn off the LED so the user knows the attack has started

keys(gui, r)
string("cmd")
keys(enter)
```

Example 4: Bypass NRO in Windows 11 This script opens a command prompt from the Windows 11 OOBE phase and executes command OOBE\BYPASSNRO, for bypassing the otherwise mandatory Internet connection and Microsoft Account requirement, to allow setting-up Windows 11 with a local user account. Note the escaped backslash in the string function call. Double quotes and backslashes need to be escaped using a backslash.

```
defaultdelay 300

delay(3000)
keys(shift, f10)
delay(1000)
keydown(alt)
key(tab)
key(tab)
keyup()
string("OOBE\\BYPASSNRO")
key(enter)
led(on)
```

Example 5: Disable OneDrive This script opens a command prompt, kills the OneDrive task, then removes the run entry from the registry for the current user, effectively disabling OneDrive for the currently logged-in user. Note the escaped backslashes and double quotes in the string function call. Double quotes and backslashes need to be escaped using a backslash. Note the use of cls and echo to communicate the intent of the script, and waitblink(1) for confirmation to proceed by pressing the button on the Digispark.

```
defaultdelay 300

delay(1000)
keys(gui, r)
string("cmd")
key(enter)
delay(1000)
string("cls && echo Press button to proceed with disabling OneDrive")
key(enter)
waitblink(1)
key(escape)
string("taskkill /f /im OneDrive.exe")
key(enter)
delay(1000)
string("reg delete \"HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run\" /v OneDrive")
key(enter)
delay(500)
key(y)
```

```
key(enter)
led(on)
```

Important Limitations

- **Keyboard Layouts:** The device uses a US English HID mapping. If the target computer uses a different layout (e.g., AZERTY or QWERTZ), certain symbols like `/`, `\`, and `@` may not map correctly.
- **Execution:** The script runs exactly once per plug-in. To repeat, unplug and re-insert the device.
- **Script Size Limitations:** The Digispark has very limited RAM and flash storage. I tested the upper limit of just single keystrokes as 824 single keys with no other commands. That yielded a 6012 byte sketch. Adding one more keystroke yields error “Flashing failed! Did the Digispark time out?”. Additionally, very long string literals will fail to work due to RAM limitations. There are mitigation strategies to limited flash storage, such as seeing if [Shift] + [Tab] might get you to a needed button/field faster than [Tab], or cutting back on other commands for only critical scripting commands.
- **Increasing String Typing Rate:** If string typing is too slow for your needs then search for line...

```
fprintf(out, "    DigiKeyboard.delay(15);\n");
```

...comment it out, and recompile the IDE program. That will remove the 15ms delay between keypresses in strings. However, if you find that keys are being missed then you should make sure that line is enabled again. Tinkering with the delay could be a workable alternative.

Fixes and Features to Possibly Add:

- Text copied and pasted into Notepad from the IDE yields a slew of odd CRLFs.
- Integrating DigiKeyboard delay amount into the IDE would be a nice feature so that tweaking that doesn't require recompiling from source.
- Add an icon for the EXE. Perhaps achievable with ResourceHacker.